

Database Design (MongoDB)

Collections Overview

- 1. Users
- 2. Outages
- 3. Notifications

4.2.1 Database Collections						
Collection Name: Users						
Description: Stores user authentication and profile details.						
Sr. No	Field Name	Datatype	Size	Description	Constraint	Example
1	user_id	ObjectId	24	Unique identifier for the user	Primary Key	64f8e2c3a5f2b1c9d8e7f01
2	name	String	100	Full name of the user	Required	Amit Patel
3	email	String	100	Email address of the user (unique)	Unique, Required	amit.patel@example.com
4	password	String	255	Encrypted password for authentication	Required	\$2b\$10\$K7v.W8k... encryptedhash
5	role	String	20	Role of the user in the system	Required	user / admin / technician
6	created_at	Date	-	Account creation timestamp	Required	2024-05-21T10:30:00Z

2. Outages Collection						
Collection Name: Outages						
Description: Stores information about reported outages.						
Sr. No	Field Name	Datatype	Size	Description	Constraint	Example
1	outage_id	ObjectId	24	Unique identifier for the outage	Primary Key	664f8e2c3a5f2b1c9d8e7f02
2	title	String	150	Title of the outage	Required	Power Cut in Sector 12
3	description	String	500	Detailed description of the outage	Required	There is a power cut in Sector 12 since morning.
4	location	String	200	Location of the outage	Required	Sector 12, Gandhinagar
5	priority	String	10	Priority level of the outage (Low / Medium / High)	Required	High (Low / Medium / High)
6	status	String	20	Current status of the outage (Pending / In Progress / Resolved)	Required	Pending (Pending / In Progress / Resolved)
7	user_id	ObjectId	24	User who reported the outage	Foreign Key	674abc90de11f1a2b3c4d5e6
8	created_at	Date	-	Outage creation timestamp	Required	2024-05-21T10:30:00Z

3. Notifications Collection

Collection Name: **Notifications**

Description: Stores alerts and updates sent to users.

Sr. No	Field Name	Datatype	Size	Description	Constraint	Example
1	notification_id	ObjectId	24	Unique identifier for the notification	Primary Key	664f8e2c3a5f2b1c9d8e7f03
2	user_id	ObjectId	24	User who receives notification	Foreign Key	674abc90de11f1a2b3c4d5e6
3	message	String	300	Notification message	Required	Your outage report has been assigned to a technician.
4	status	String	10	Read status of notification	Required	Unread (Read / Unread)
5	created_at	Date	–	Notification creation timestamp	Required	2024-05-21T11:00:00Z